

Data Formats: XML & JSON

Week 4 · Foundations module · Textbook Chapter 4

Brian C. Keegan, Ph.D.

Associate Professor, Department of Information Science
University of Colorado Boulder

September 9 & 11, 2026

Two formats carry almost all web data: **XML** (a tree of tags) and **JSON** (nested lists and dictionaries).

Monday | Labor Day — no class

- Monday's concept intro is folded into Wednesday

Wednesday | Concepts & notebook lab

- XML & JSON concepts, then the Chapter 4 notebook: Congress, tweets, and weather

Friday | Textbook revisions

- Propose & peer-review pull requests from Chapters 1–4

Companion notebook

`ch-04-data-formats.ipynb`

- **What happens if an issue is resolved incorrectly?** — GitHub is primarily a version control system, so it is possible to “rewind” to an earlier version. In other words, don’t stress you’ll “break” something.
- **Don’t understand how to file a pull request.** — We’ll keep practicing more this Friday, I need to do more scaffold a good example.
- **Concerns with citing information added to textbook?** — You should all have “write” access to merge pull requests right away, which will update the textbook for everyone.
- **Will all issues eventually be merged?** — Probably not, but that isn’t what you’re graded on.

1. Wednesday · Concepts & Notebook

Two formats dominate data from the web:

- **XML** — the older format and a down-payment on HTML: a tree of nested tags.
- **JSON** — the newer, lighter format of the API economy: lists and dictionaries.

Every page, API response, and data feed you meet this semester is going to be a flavor of XML or JSON

When to use which

XML: older web services, government feeds (Congress, SEC, RSS), configuration files.

JSON: modern web APIs, database exports, service-to-service exchange.

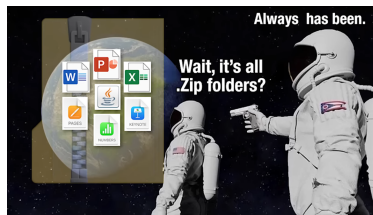
A brief history of XML

XML and HTML are often assumed to be one a subset of the other, but they are siblings descending from SGML

The HTML you scrape in the wild almost never is valid XML, need a different more generous parser

XML is a foundation for a whole family of other languages: XHTML (websites), SVG (images), MathML (math equations), RSS (podcasts)

XML hides in plain sight: open a `.docx`, `.xlsx`, or `.svg` file's internals and you are looking at it.



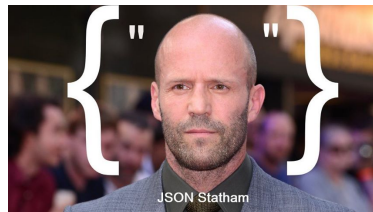
A brief history of JSON

JSON specified in the early 2000s

JSON's design goal was the opposite of XML's: no tags, no schema, no comments — “just the data”

Objects and arrays that can be recognized by all modern programming languages and easy for machines to parse

As Python-native coders, JSON intuitively matches lists and dictionaries you already know



XML: a tree of nested tags

XML organizes data as a **hierarchy**: each tag can hold text, attributes, and child tags. The House MemberData feed, drawn as a tree:

```
<MemberData>
  <member> # repeats 441 times
    <member-info>
      <firstname>Joe</firstname> # value in TEXT
      <lastname>Neguse</lastname>
      <party>D</party>
    </member-info>
    <state postal-code="CO"> # value in an ATTRIBUTE
      <state-fullname>Colorado</state-fullname>
    </state>
  </member>
</MemberData>
```

A value hides in one of two places: **text** (read with .text) or an **attribute** (read with brackets).

JSON: lists and dictionaries you already know

JSON maps onto two Python structures you already use. The weather forecast, peeled one layer at a time:

```
{ # dict: type(data) is dict
  "latitude": 40.01,
  "daily": { # dict: data["daily"]
    "time": [...], # list: data["daily"]["time"]
    "temperature_2m_max": [...]
  }
}
```

A JSON **array** is a Python **list**; a JSON **object** is a Python **dict**. Real data nests them several levels deep — the skill is navigating from the outside in, one `type()` and `.keys()` call at a time.

JSON vs. XML: pick your trade-offs

JSON

- + Compact — no repeated tag names
- + Parses natively in JavaScript
- + Simple: objects, arrays, 4 scalar types
- No comments allowed
- No native schema enforcement
- No display semantics — data only

XML

- + Schema validation (DTD/XSD) before parsing
- + Rich vocabulary: attributes, mixed content, namespaces
- + Can double as a document format (SVG, DOCX)
- Verbose — every tag closes
- No native browser/JS parsing
- Everything parses as text; you convert types

Neither is intrinsically better than the other — just different tools for different needs

Where JSON and XML show up

JSON dominates anywhere a **program** is the primary reader, particularly APIs

XML persists where a human sometimes needs to read the document, where a strict contract matters, or where the data predates the JSON era.

Government and legislative systems still commonly use XML — partly because XSD validation guarantees a submitted document has the fields it is supposed to have.

Rule of thumb

If a program wrote it and a program will read it, expect JSON.

If humans are involved in read/write or high-stakes data, expect XML.

Whether data starts as XML, JSON, HTML, or PDF text, the move into analysis is always the same:

find records → **extract fields** → **lists + dicts** → **pd.DataFrame**

Once you are in a DataFrame, a whole host of other data science methods for cleaning, filtering, grouping, aggregating, merging open up

Parsing XML with BeautifulSoup

Parse the string into a navigable tree, then search it. `.find()` returns the first match, `.find_all()` returns them all — each searches the *entire* subtree beneath a tag:

```
from bs4 import BeautifulSoup

soup = BeautifulSoup(house_raw, "xml") # the "xml" parser comes from lxml

members = soup.find_all("member")
print(len(members)) # 2

for m in members:
    first = m.find("firstname").text # .text = content between the tags
    last = m.find("lastname").text # .find() reaches nested tags too
    party = m.find("party").text
    print(f"{first} {last} ({party})") # Joe Neguse (D) / Jeff Hurd (R)
```

Data exists in two different places in XML

A value is either **text** or an **attribute**:

```
state = soup.find("state")
state["postal-code"] # "CO"
state.find("state-fullname").text # "Colorado"
```

A missing tag makes `.find()` return `None`:

```
# AttributeError: None has no .text
m.find("party").text
# guard every extraction instead:
m.find("party").text if m.find("party") else None
```

Working out *which* of the two holds your value is half the parsing battle — real XML uses both.

Calling `.text` on a `None` is the **single most common** scraping error. The `if ...else None` guard keeps one missing field from crashing a 441-record loop.

XML always hands you **text**: converting "2nd" or "\$1,234" to a number is your job.

XML → DataFrame: the real House roster

```
import requests, pandas as pd

url = "https://clerk.house.gov/xml/lists/MemberData.xml"
soup = BeautifulSoup(requests.get(url).content, "xml")
print(soup.find("MemberData")["publish-date"])

records = []
for m in soup.find_all("member"): # 441 members
    st = m.find("state")
    records.append({"last_name": m.find("lastname").text,
                    "party": m.find("party").text,
                    "state": st["postal-code"] if st else
                        None})
df = pd.DataFrame(records) # find_all -> dicts ->
    DataFrame
```

The raw response, before parsing:

July 1, 2026

```
<member>
  <member-info>
    <firstname>Joe</firstname>
    <lastname>Neguse</lastname>
    <party>D</party>
  </member-info>
  <state postal-code="CO">
    <state-fullname>
      Colorado
    </state-fullname>
  </state>
</member>
```

JSON: parse a string, then the list-of-dictionaries pipeline

`json.loads()` turns a string into Python objects (`json.load()` reads a file). With `requests`, `response.json()` does it for you:

```
import json, pandas as pd

data = json.loads('{ "state": "Colorado", "population": 5773714 }')
print(data["state"]) # Colorado
# response.json() == json.loads(response.text)

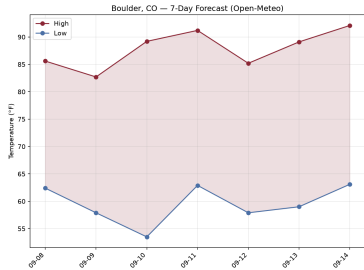
census = [ # the list-of-dictionaries pattern:
    { "state": "Colorado", "year": 2020, "population": 5773714 },
    { "state": "Colorado", "year": 2010, "population": 5029196 },
]
df = pd.DataFrame(census) # one dict per row, keys -> columns
```

JSON: peel the onion on a live API

```
import requests

url = "https://api.open-meteo.com/v1/forecast"
params = {"latitude": 40.01, "longitude": -105.27,
          "daily": "temperature_2m_max",
          "timezone": "America/Denver"}
data = requests.get(url, params=params).json()

# type() + .keys(), one level deeper each time
data.keys() # latitude, longitude, daily...
data["daily"].keys() # time, temperature_2m_max
highs = data["daily"]["temperature_2m_max"] # a list
data.get("elevation", "unknown") # .get avoids KeyError
```



Boulder's 7-day forecast, fetched live on Sept. 8.

Lab: work these in pairs, then show-and-tell

Work in pairs. Do these exercises from the Chapter 4 notebook:

1. **RSS feed** — parse a news feed with the "xml" parser. Extract the title, the link, and the date. Put them in a DataFrame. Starter feeds and how to find your own: this week's handout.
2. **Nested JSON** — get the dates and the maximum temperatures from Open-Meteo. Put them in a DataFrame.
3. **XML and JSON** — get the same data in both formats. Report which format is easier to parse. Report which format has more metadata.
4. **Reusable** — write `xml_to_dataframe(soup, tag, fields)`. Test it on the Congress data. Then test it on an RSS feed.
5. **Weather** — plot the high and low temperatures. Mark the days below freezing.
6. **5617**: write a critique of a public XML or JSON dataset. Use the datasheet format of Gebru et al. (2021)

Show-and-Tell

Bring one of these to class:

- a bug that you could not solve
- data that you collected and find interesting
- a question for a research design

Error flavors

Some code here is *designed* to break (a guard demonstrating a crash) — that does not need a GitHub issue. Other code may be *genuinely* broken: confirm it, then search and file an issue.

Effort over perfection

Start the lab in class, finish at home, submit what you got working. Graded on a good-faith attempt to complete the notebook, not on getting every cell to run.

Submitting

Run all cells, then export: *File* → *Save and Export Notebook As* → *HTML*. Upload to Canvas by **Sunday 11:59pm**.

2. Friday · Textbook Revisions

A worked example — linking the Missing Manual

Until this week, `ch-04-data-formats.qmd` had this line:

“For more on data file formats... see Missing Manual Chapter 20: Data File Formats.”

It named a chapter. It didn't link to it. A reader had to leave the page, find the Missing Manual site, and search for Chapter 20 by hand.

Here's exactly how that got fixed — eight steps, start to finish. Yours will be a different gap; the mechanics are identical.

The target

```
https://cuinfoscience.github.io/  
INFO-Missing-Manual/chapters/  
data-file-formats.html
```

Step 1 — find the file

Two ways in:

1. Open the rendered chapter. Click **Edit this page** in the sidebar.
2. Or browse the repo directly:
`cuinfoscience/Web-Data-Science-Book`, then open
`ch-04-data-formats.qmd`.

Both paths land on the same page: the raw `.qmd` source, in GitHub's file viewer.

Edit the right file

The `.qmd` is the source. Never the rendered HTML, and never the `.ipynb` — it is generated from the chapter, and your edit would be overwritten.

Step 2 — open the editor

Click the pencil icon in the top right of the file view — **Edit this file**.

You have write access now. GitHub takes you straight into an in-browser text editor. No fork, no separate copy — you are editing the real repository.

One change per PR

A PR is a conversation about one change. Add one link today, not everything you happen to notice.

Step 3 — find the line

The chapter runs past 500 lines. Do not scroll to find one sentence.

Click inside the editor. Press **Ctrl+F** (**Cmd+F** on a Mac) to open the editor's own search bar.

Search for `Missing Manual Reference`. It appears once, near the end of the chapter.

The wrong search bar still works, badly

Your browser's Ctrl+F only searches what is currently rendered. GitHub's editor holds the whole file. Search inside the editor pane, not the page.

Step 4 — write the Markdown link

A Markdown link has the form [text] (url). Wrap the existing phrase — do not retype it:

Before:

```
...see *Missing Manual* Chapter 20: Data File Formats.
```

After:

```
...see [*Missing Manual* Chapter 20: Data File Formats]  
(https://cuinfoscience.github.io/INFO-Missing-Manual/  
chapters/data-file-formats.html).
```

The italics markup stays inside the brackets. Link text can carry other formatting.

Click the **Preview** tab above the editor.

Check three things:

1. The phrase is blue and underlined — a live link.
2. The italics survived — *Missing Manual* still reads in italics.
3. Nothing else on the page changed.

Do not click the link yet

Verify it visually first. Test that it actually resolves after the PR's render check runs.

Step 6 — commit to a new branch

Scroll down. Below the editor, GitHub asks how to save your change.

Write a short commit title: *“Link the Missing Manual chapter reference in Ch. 4.”*

Select **Create a new branch for this commit and start a pull request** — not *Commit directly to the main branch*.

Write access is not a shortcut

You can push straight to `main`. That does not mean you should. A branch gives a reviewer something to read before it ships.

Step 7 — open the pull request

Click **Propose changes**.

GitHub moves you to the **Comparing changes** screen: your branch on the left, main on the right, the diff in between.

Check the diff. One line removed, one line added, nothing else. That is what “one change per PR” looks like on screen, not just in principle.

Click **Create pull request** to open the title and description form.

Step 8 — write a good description

Use the same four fields as your issues, now describing a fix instead of a problem:

Location: `ch-04-data-formats.qmd`, "Missing Manual Reference"

Problem: The callout names Chapter 20 but gives no way to click through to it.

Why: A reader has to leave the page and search the Missing Manual site by hand to find the chapter.

Change: Wrapped the existing text in a Markdown link to the chapter's page.

Click **Create pull request**. Your PR is open.

What happens next

Every PR against the book triggers a render check. Green means the book still builds with your change. Red means read the log before asking — the failing line is usually named.

Watch for a reviewer's comments. Push a follow-up commit to the same branch if asked, or defend your choice.

When it merges, the book republishes itself. Reload the chapter in a few minutes — your fix is live, the same way this one is.

The loop you're about to close

Reader → found a gap → edited → opened a PR → merged → live page. All of it in a browser tab.

High-value targets — pick one and scope it to a single PR:

- **Test the code against live sources.** The House roster drifts (441 today; the `publish-date` changes) and Open-Meteo dates roll forward. Run the notebook, report what changed, and pin the snapshot.
- **Close a gap in “Common Issues to Debug.”** Add a worked `json.JSONDecodeError` case — a trailing comma, single quotes, or an HTML error page returned instead of JSON.
- **Verify a real-world-patterns claim.** The chapter names specific places JSON and XML show up (JWTs, SOAP, Office file formats). Check one against a primary source and tighten the citation.
- **Strengthen an exercise.** Give the RSS or weather exercise expected output, a rubric, or a starter cell so it is self-checking.
- **Extend the RSS / post-API story.** Tie the decline-of-RSS narrative to a concrete citation and the “soft enclosure” theme (Ch. 3).

A good revision, and a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Renders (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates “must fix” from “nice to have”
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Web Architecture & Protocols** — how a URL becomes a rendered page (HTTP, requests, status codes).

Reviewing someone else's PR? Do not just describe the fix in a comment. Propose it directly:

```
'''suggestion
XML 1.0 became a W3C
Recommendation in 1998.
'''
```

One click for the author to accept it. This is the fastest way to turn “must fix” into a merged change.

Key takeaways

1. **XML** is a tree of tags; **JSON** is nested lists and dictionaries — the two structural languages of web data.
2. XML (1998, from SGML) favors strict contracts and documents a human might read; JSON (early 2000s, from JavaScript) favors compact, program-to-program exchange.
3. BeautifulSoup navigates XML with `.find()`, `.find_all()`, and `.text`; attributes come out with bracket notation. For JSON, **peel the onion** — `type()` and `.keys()`, top-down.
4. The `find_all` → list-of-dicts → `DataFrame` pipeline is the bridge from raw web data to analysis.
5. Format is a **transparency** choice: open XML and JSON support public oversight; scanned PDFs prevent it.